

EX.NO:4 CREATE AN ABSTRACT CLASS-SHAPE CLASS

Program:

```
import java.util.Scanner;

abstract class LibraryMember {
    int memberId;
    String name, email, phone;

    LibraryMember(int memberId, String name, String email, String phone) {
        this.memberId = memberId;
        this.name = name;
        this.email = email;
        this.phone = phone;
    }

    void display() {
        System.out.println("Member ID : " + memberId);
        System.out.println("Name      : " + name);
        System.out.println("Email   : " + email);
        System.out.println("Phone   : " + phone);
    }

    abstract void generateSummary();
}

class StudentMember extends LibraryMember {

    StudentMember(int id, String name, String email, String phone) {
        super(id, name, email, phone);
    }

    void generateSummary() {
        System.out.println("\n----- Student Member -----");
        display();
        System.out.println("Borrowing Limit      : 5 Books");
        System.out.println("Penalty Per Day      : Rs.2");
        System.out.println("Annual Membership Fee: Rs.200");
    }
}
```

```

class FacultyMember extends LibraryMember {

    FacultyMember(int id, String name, String email, String phone) {
        super(id, name, email, phone);
    }

    void generateSummary() {
        System.out.println("\n----- Faculty Member -----");
        display();
        System.out.println("Borrowing Limit    : 10 Books");
        System.out.println("Penalty Per Day      : Rs.3");
        System.out.println("Annual Membership Fee: Rs.500");
    }
}

class ExternalMember extends LibraryMember {

    ExternalMember(int id, String name, String email, String phone) {
        super(id, name, email, phone);
    }

    void generateSummary() {
        System.out.println("\n----- External Member -----");
        display();
        System.out.println("Borrowing Limit    : 3 Books");
        System.out.println("Penalty Per Day      : Rs.5");
        System.out.println("Annual Membership Fee: Rs.1000");
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Member ID: ");
        int id = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Email: ");
        String email = sc.nextLine();
        System.out.print("Enter Phone: ");
    }
}

```

```

String phone = sc.nextLine();

System.out.println("\nSelect Member Type");
System.out.println("1. Student");
System.out.println("2. Faculty");
System.out.println("3. External");
System.out.print("Enter Choice: ");
int choice = sc.nextInt();

switch (choice) {
    case 1:
        StudentMember s = new StudentMember(id, name, email, phone);
        s.generateSummary();
        break;
    case 2:
        FacultyMember f = new FacultyMember(id, name, email, phone);
        f.generateSummary();
        break;
    case 3:
        ExternalMember e = new ExternalMember(id, name, email, phone);
        e.generateSummary();
        break;
    default:
        System.out.println("Invalid Choice");
}
sc.close();
}
}

```

Output:

```

Enter Member ID: 101
Enter Name: Janaki
Enter Email: janaki@gmail.com
Enter Phone: 9876543210

```

```

Select Member Type
1. Student
2. Faculty
3. External
Enter Choice: 1

```

----- Student Member -----

Member ID : 101

Name : Janaki

Email : janaki@gmail.com

Phone : 9876543210

Borrowing Limit : 5 Books

Penalty Per Day : Rs.2

Annual Membership Fee: Rs.200